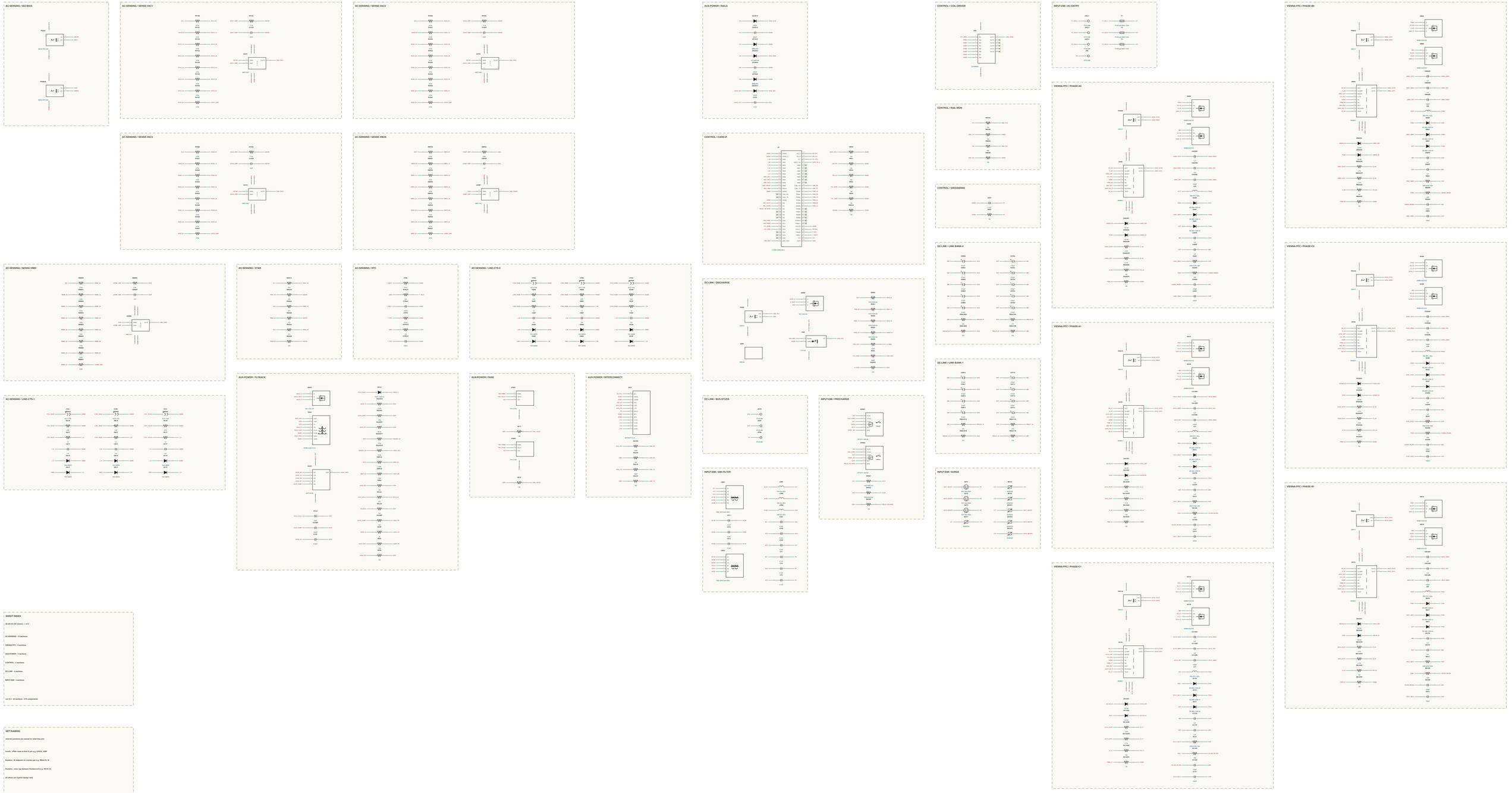
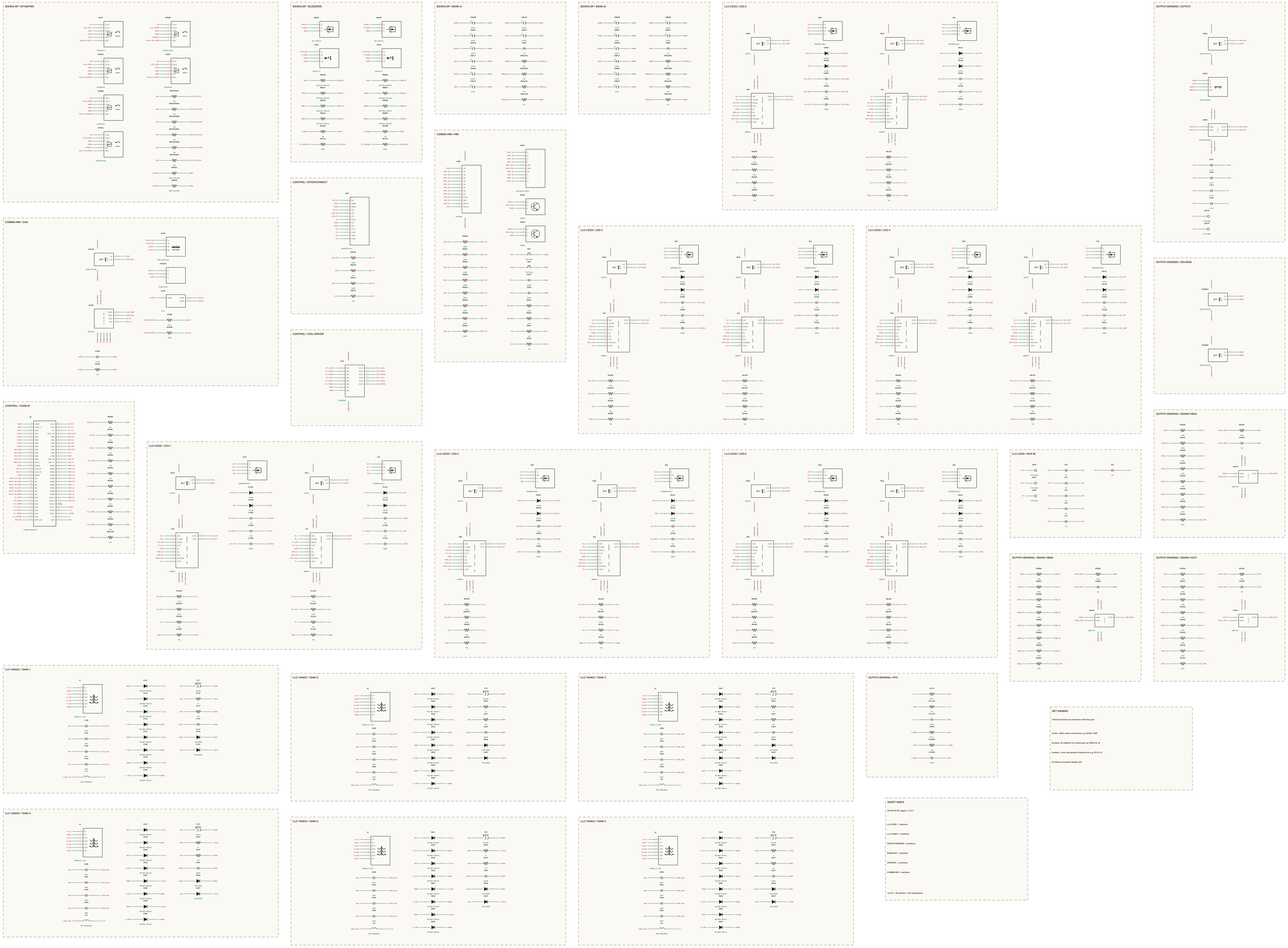
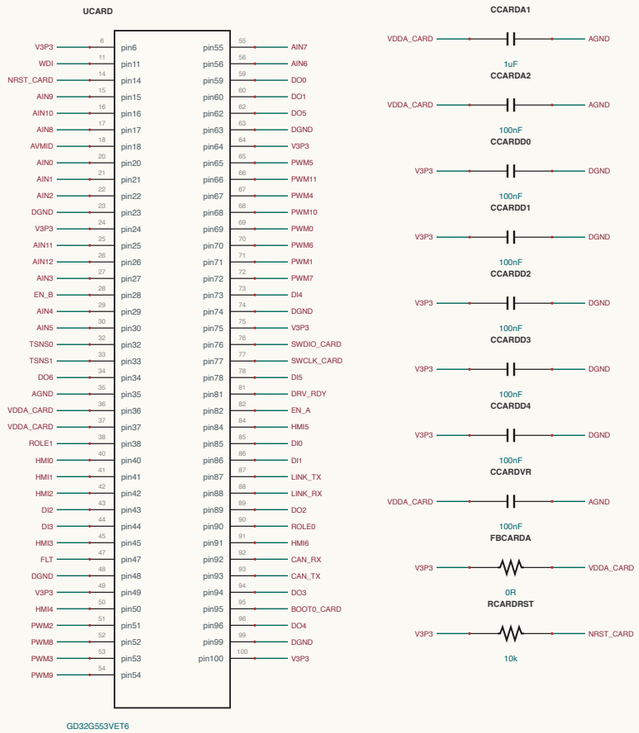
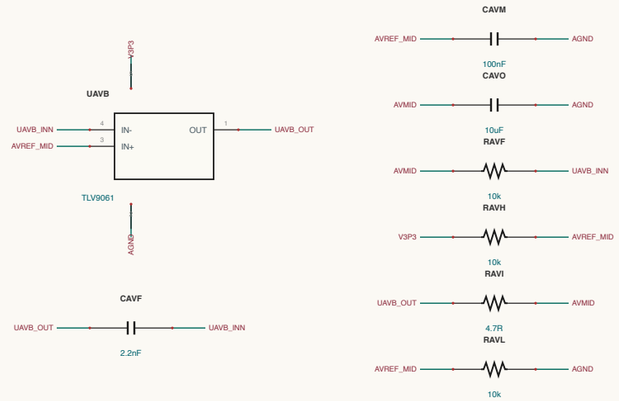

AC-SENSING / ISO-BIAS
AC-SENSING / SENSE-VAC1
AC-SENSING / SENSE-VAC2
AUX-POWER / RAIL5
CONTROL / COIL-DRIVER
INPUT-EMI / AC-ENTRY
VIENNA-PFC / PHASE-B0
AC-SENSING / SENSE-VAC3
AC-SENSING / SENSE-VBUS
CONTROL / RAIL-MON
VIENNA-PFC / PHASE-A2
CONTROL / CARD-IF
CONTROL / GROUNDING
VIENNA-PFC / PHASE-C0
AC-SENSING / SENSE-VMID
AC-SENSING / STAR
AC-SENSING / NTC
AC-SENSING / LINE-CTS-0
DC-LINK / LINK-BANK-0
DC-LINK / DISCHARGE
VIENNA-PFC / PHASE-A1
VIENNA-PFC / PHASE-B3
AC-SENSING / LINE-CTS-1
AUX-POWER / FLYBACK
AUX-POWER / FANS
AUX-POWER / INTERCONNECT
DC-LINK / LINK-BANK-1
DC-LINK / BUS-STUDS
INPUT-EMI / PRECHARGE
INPUT-EMI / EMI-FILTER
INPUT-EMI / SURGE
VIENNA-PFC / PHASE-C1
SHEET INDEX
DC-Modules 60 kW — AC-DC board (Vienna PFC, 2x cells)
DC-Modules 60 kW - board AC-DC (lower), sheet 1 of 3
2026-09-06 · rev D.3
NET NAMING


BANKS-SP / SP-MATRIX
BANKS-SP / BLEEDERS
BANKS-SP / BANK-A
BANKS-SP / BANK-B
LLC-LEGS / LEG-2
OUTPUT-SENSING / OUTPUT
COMMS-HMI / CAN
CONTROL / INTERCONNECT
COMMS-HMI / HMI
CONTROL / COIL-DRIVER
OUTPUT-SENSING / ISO-BIAS
LLC-LEGS / LEG-3
LLC-LEGS / LEG-4
CONTROL / CARD-IF
LLC-LEGS / LEG-1
OUTPUT-SENSING / SENSE-VBKA
LLC-LEGS / LEG-5
LLC-LEGS / LEG-6
LLC-LEGS / BUS-IN
OUTPUT-SENSING / SENSE-VBKB
OUTPUT-SENSING / SENSE-VOUT
LLC-TANKS / TANK-1
LLC-TANKS / TANK-2
LLC-TANKS / TANK-3
OUTPUT-SENSING / NTC
NET NAMING
LLC-TANKS / TANK-4
LLC-TANKS / TANK-5
LLC-TANKS / TANK-6
SHEET INDEX
DC-Modules 60 kW — DC-DC board (3-phase LLC, 2x cells)
DC-Modules 60 kW - board DC-DC (upper), sheet 2 of 3
2026-09-06 · rev D.3

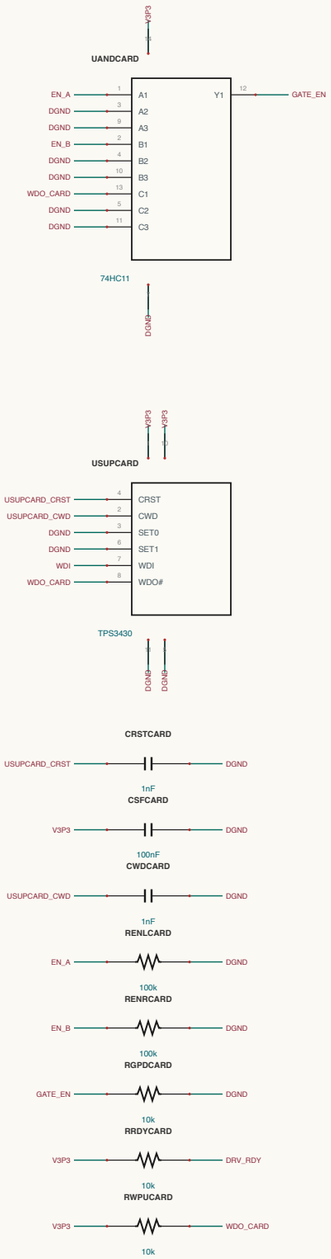
CONTROL / MCU



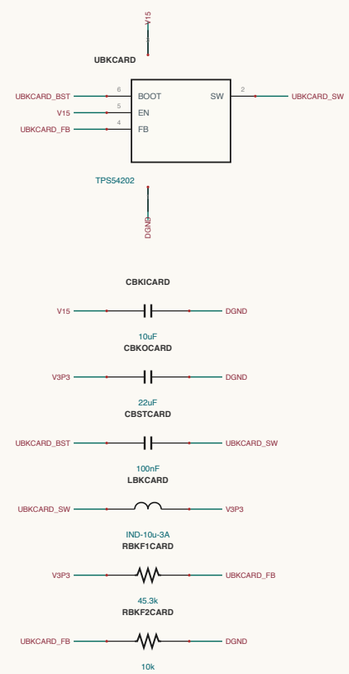
CONTROL / ANALOG-MID



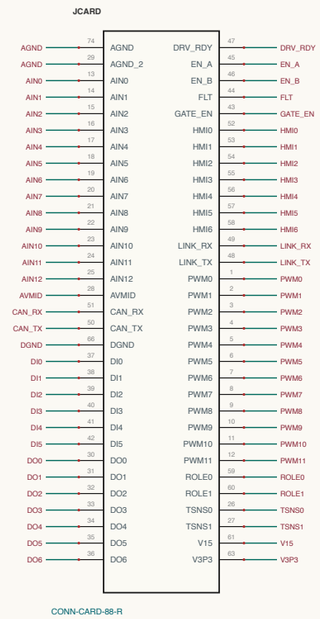
CONTROL / SAFETY



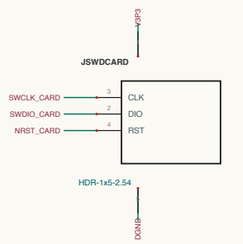
CONTROL / BUCK-3V3



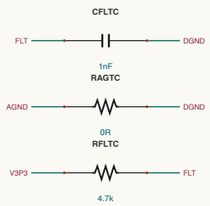
CONTROL / CARD-IF



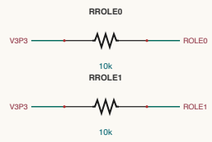
CONTROL / SWD-BOOT



CONTROL / FLT-GROUND



CONTROL / ROLE



30/60 kW Control card - 3 of 3 · rev D.3 · 8 sections · 45 components

NET NAMING: U<ref>_<PIN> = node at that IC pin · R<stem>_M = series-pair midpoint · R<stem>_<nm> = tap between R<stem>n/m · all others are explicit design nets

DC-Modules — Control Card (GD32G553VET6, one card for both converter roles)

DC-Modules 30/60 kW - board Control card, sheet 3 of 3

2026-09-06 · rev D.3